

CSE/ISyE 6740
Computational Data Analysis

Dimensionality Reduction

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Outline

- **Unsupervised Learning**
 - **Linear dimensionality reduction**
 - Principal component analysis
 - Eigenvalue decomposition
 - Reconstruction
 - **Non-linear dimensionality reduction**
 - Isomap
 - How Isomap works?
 - Other non-linear dimensionality reduction techniques

Matrix and Vector Convention

- A data point (feature) is always a column vector in $\mathbb{R}^d$, with dimensionality d

$$x^i = \begin{bmatrix} x_1^i \\ x_2^i \\ \vdots \\ x_d^i \end{bmatrix} \in \mathbb{R}^d$$

- Feature matrix is a $n \times d$ matrix, concatenating n data points (transposed)

$$X = [x^1, x^2, \dots, x^n]^\top = \begin{bmatrix} x^{1\top} \\ x^{2\top} \\ \vdots \\ x^{n\top} \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_d^1 \\ x_1^2 & x_2^2 & \dots & x_d^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^n & x_2^n & \dots & x_d^n \end{bmatrix}$$

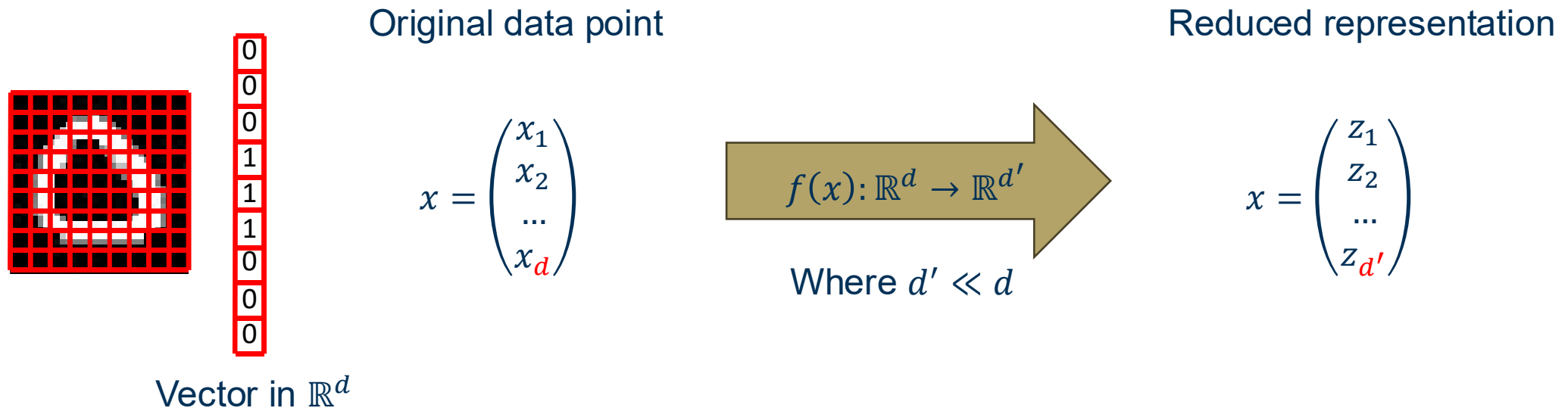
Data point #1

Feature #2

Dimensionality Reduction

What is Dimensionality Reduction?

- The process of reducing the number of random variables under consideration
 - One can combine, transform or select variables
 - One can use linear or nonlinear operations



Why Dimensionality Reduction and How to Think?

- The dimension-reduced data can be used for
 - Visualization
 - Aggregating weak signals in the data
 - Cleaning the data
 - Speeding up subsequent learning task
 - Simplify model
- Key questions of a dimensionality reduction algorithm
 - What is the criterion for carrying out the reduction process?
 - What are the algorithm steps?

Principal Component Analysis

- Given n data points, $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- **Step 1:** estimate the mean and covariance matrix from the data:
 - $\mu = \frac{1}{n} \sum_{i=1}^n x^i$ and $C = \frac{1}{n} \sum_{i=1}^n (x^i - \mu)(x^i - \mu)^\top$
- **Step 2:** take the eigenvectors $w^1, w^2, \dots$ of C corresponding to the largest eigenvalue λ_1 , the second largest eigenvalue $\lambda_2, \dots$
- **Step 3:** compute reduced representation

$$z^i = \begin{pmatrix} w^1{}^\top (x^i - \mu) / \sqrt{\lambda_1} \\ w^2{}^\top (x^i - \mu) / \sqrt{\lambda_2} \\ \dots \\ w^{d'}{}^\top (x^i - \mu) / \sqrt{\lambda_{d'}} \end{pmatrix}$$

Use What Criterion for Reduction?

- There are many criteria (geometric based, information theory based, etc.)
- **One criterion:** want to capture **variation** in data
 - Variations are “signals” or information in the data
 - Need to normalize each variables first
- In the process, also discover variables or dimensions highly **correlated**
 - Represent highly related phenomena
 - Combine them to form a stronger signal
 - Lead to simpler presentation

How to Formulate the Problem?

- Given n data points, $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$, with their mean $\mu = \frac{1}{n} \sum_{i=1}^n x^i$
- Find a direction $w \in \mathbb{R}^d$, where $\|w\| = 1$
- Such that the variance (or variation) of the data along direction w is maximized

$$\max_{w: \|w\|=1} \underbrace{\frac{1}{n} \sum_{i=1}^n (w^\top x^i - w^\top \mu)^2}_{\text{variance}}$$

Is it an Easy Optimization Problem?

- Manipulate the objective with linear algebra

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n (w^\top x^i - w^\top \mu)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (w^\top (x^i - \mu))^2 \\ &= \frac{1}{n} \sum_{i=1}^n w^\top (x^i - \mu) (x^i - \mu)^\top w \\ &= w^\top \underbrace{\left(\frac{1}{n} \sum_{i=1}^n (x^i - \mu) (x^i - \mu)^\top \right)}_{\text{Covariance matrix}} w \end{aligned}$$

Landscape of the Optimization Problem

- Suppose the data has two dimension ($d = 2$)
- C is a diagonal matrix

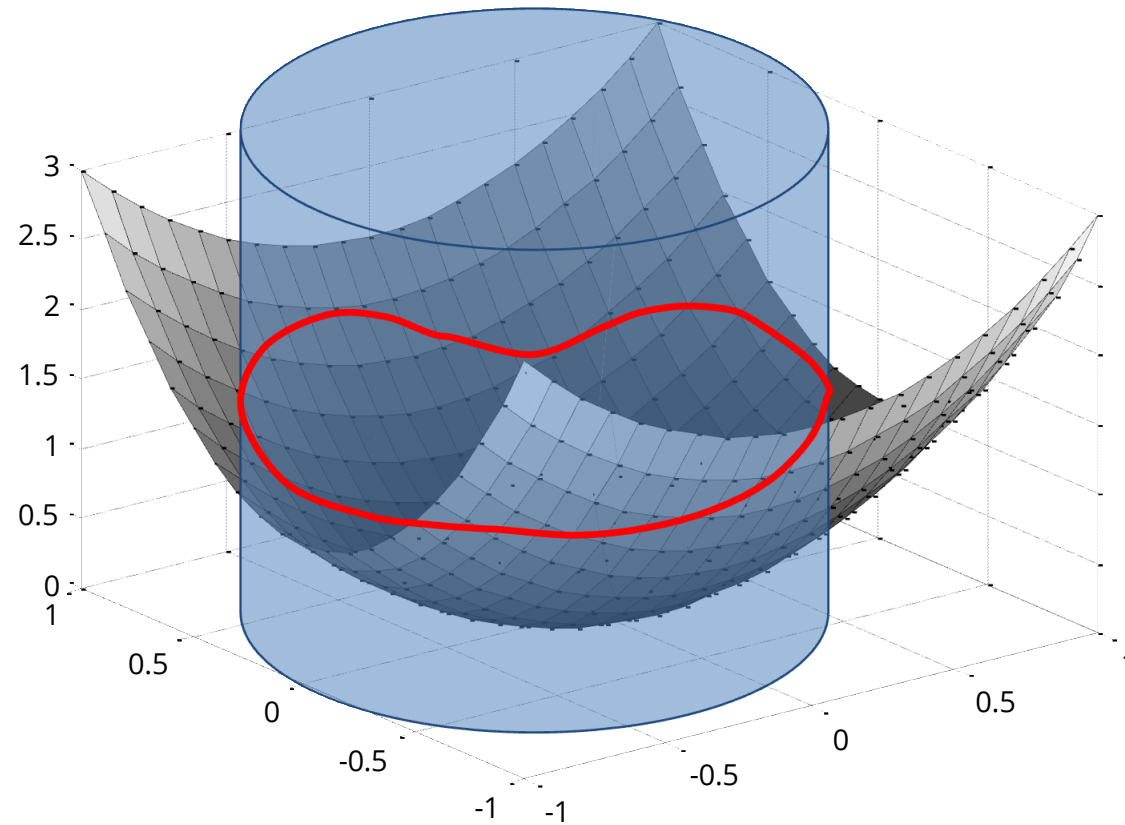
$$C = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

- The optimization problem becomes

$$\begin{aligned} & \max_{w: \|w\|=1} w^T C w \\ &= \max_{w: \|w\|=1} (w_1, w_2) \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ &= \max_{w: \|w\|=1} w_1^2 + 2w_2^2 \end{aligned}$$

Landscape of the Optimization Problem

$$f(w_1, w_2) = w_1^2 + 2w_2^2$$



Eigenvalue Problem

- **Eigenvalue problem**

- Given a symmetric matrix $C \in \mathbb{R}^{d \times d}$
- Find a vector $w \in \mathbb{R}^d$ and $\|w\| = 1$
- Such that

$$Cw = \lambda w$$

- There will be multiple solution of $w^1, w^2, \dots, w^d$ with different $\lambda_1, \lambda_2, \dots, \lambda_d$
- They are orthonormal: $w^{i\top} w^i = 1, w^{i\top} w^j = 0$

Equivalent to Eigenvalue Problem

- **Claim:**

$$\max_{w: \|w\|=1} w^T C w \Rightarrow C w = \lambda w$$

- **Proof:** Form the Lagrangian function of the optimization problem

$$L(w, \lambda) = w^T C w + \lambda(1 - \|w\|^2)$$

Necessary
condition

- If w is a maximum of the original optimization problem, then there exists a λ , where (w, λ) is a **stationary point** of $L(w, \lambda)$.

- This implies that:

$$0 = \frac{\partial L}{\partial w} = 2Cw - 2\lambda w$$

Variance in the Principal Direction

- Principal direction w satisfies

$$Cw = \lambda w$$

- Variance in principal direction is

$$\begin{aligned} w^T C w \\ &= \lambda w^T w \\ &= \lambda \end{aligned}$$

Eigenvalue

Multiple Principal Directions

- Direction $w^1, w^2, \dots, w^d$, which has
 - the largest variances
 - but are also **orthogonal** to each other
- Take the eigenvectors $w^1, w^2, \dots, w^d$ of C corresponding to
 - The largest eigenvalue λ_1 , the second largest eigenvalue λ_2 , ...

Principal Component Analysis (Revisit)

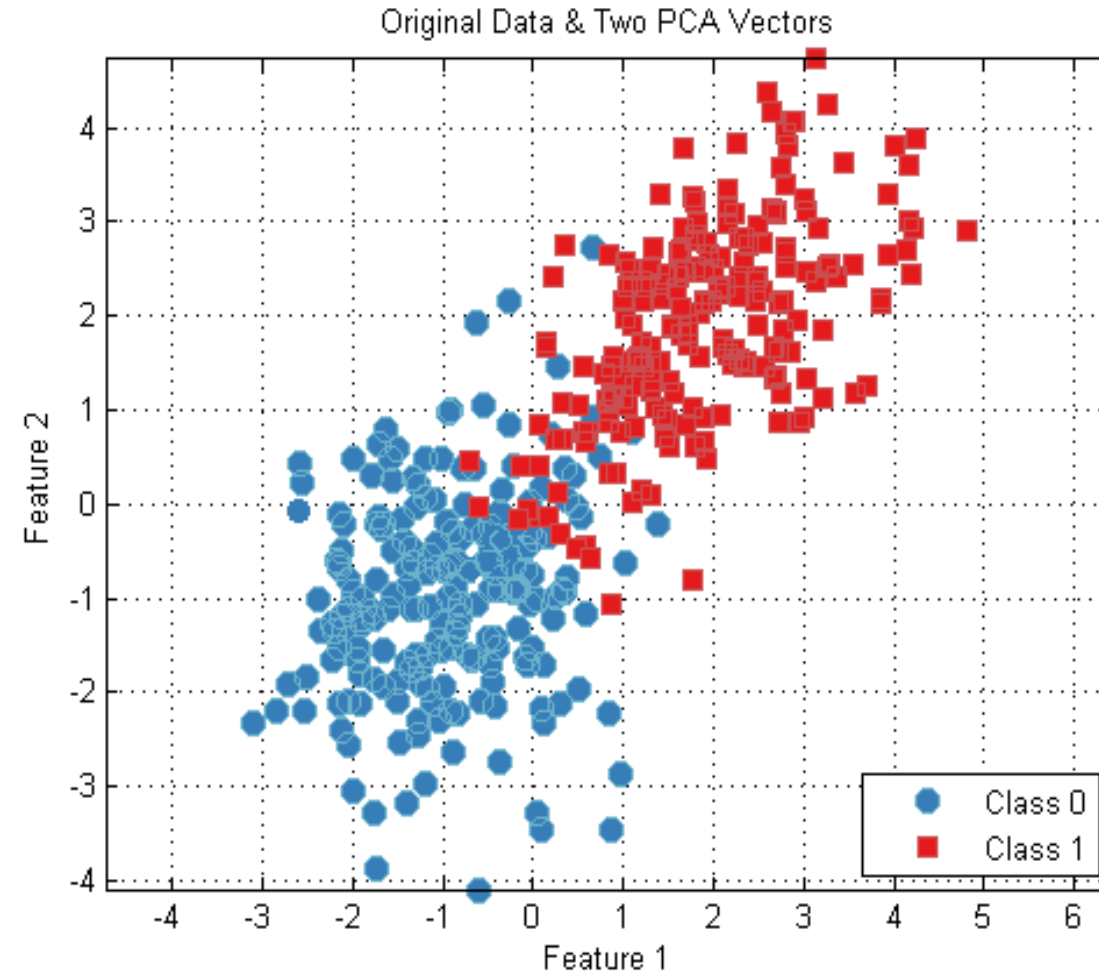
- Given m data points, $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- **Step 1:** estimate the mean and covariance matrix from the data:
 - $\mu = \frac{1}{n} \sum_{i=1}^n x^i$ and $C = \frac{1}{n} \sum_{i=1}^n (x^i - \mu)(x^i - \mu)^\top$
- **Step 2:** take the eigenvectors $w^1, w^2, \dots, w^{d'}$ of C corresponding to the largest eigenvalue λ_1 , the second largest eigenvalue $\lambda_2, \dots$
- **Step 3:** compute reduced representation

Principal directions

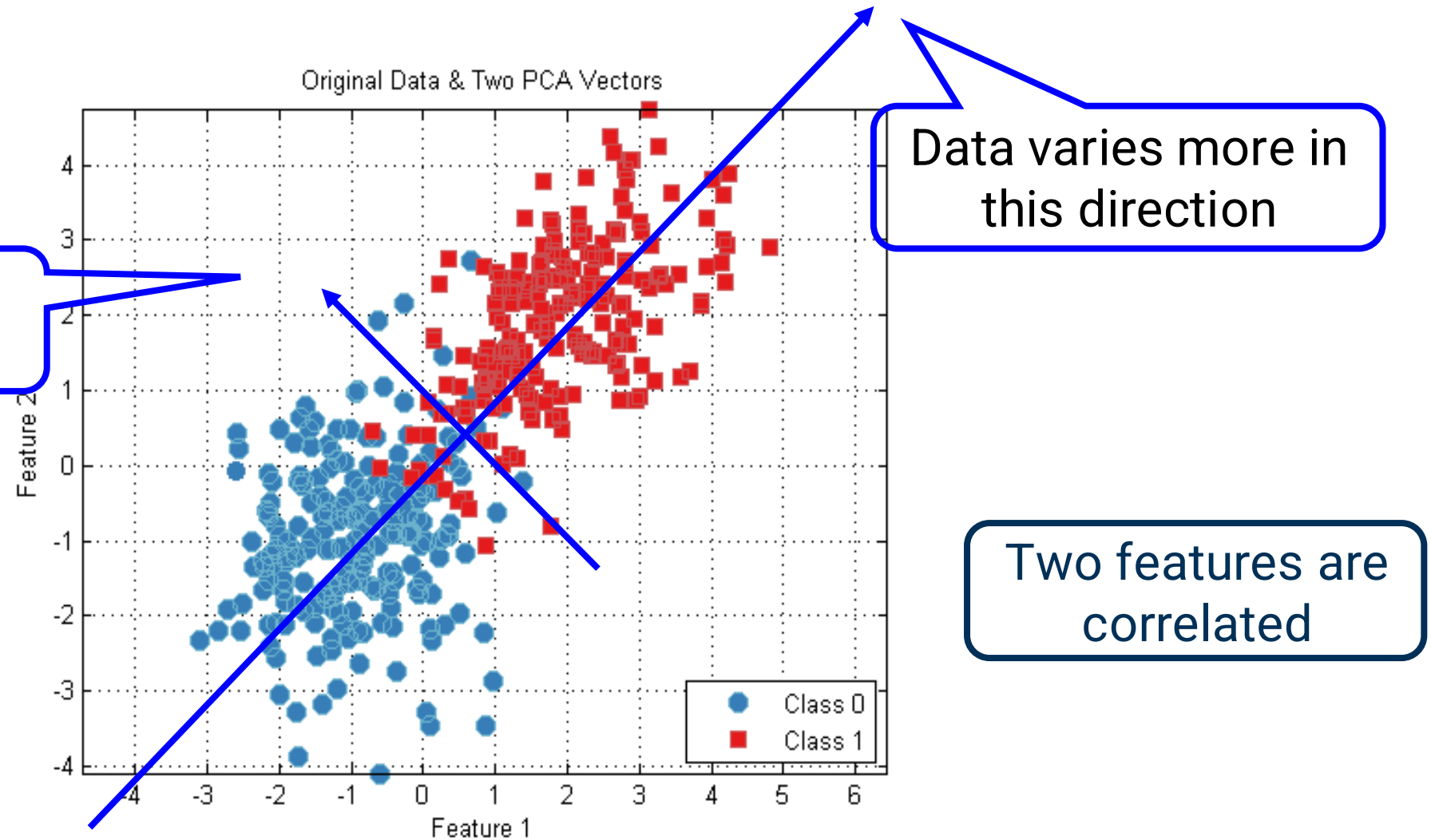
$$z^i = \begin{pmatrix} w^1{}^\top (x^i - \mu) / \sqrt{\lambda_1} \\ w^2{}^\top (x^i - \mu) / \sqrt{\lambda_2} \\ \dots \\ w^{d'}{}^\top (x^i - \mu) / \sqrt{\lambda_{d'}} \end{pmatrix}$$

Normalize by
standard deviation

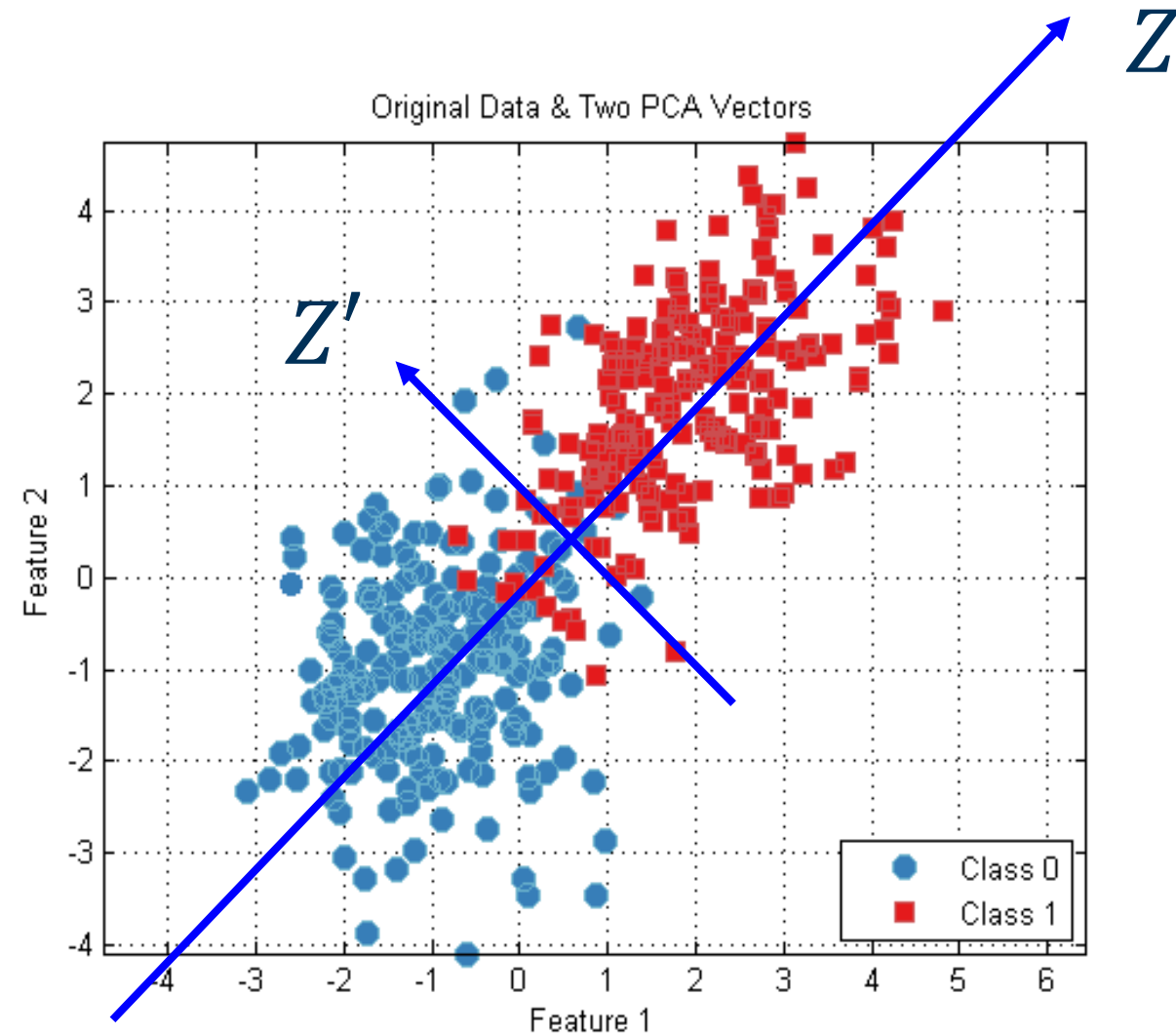
An Example



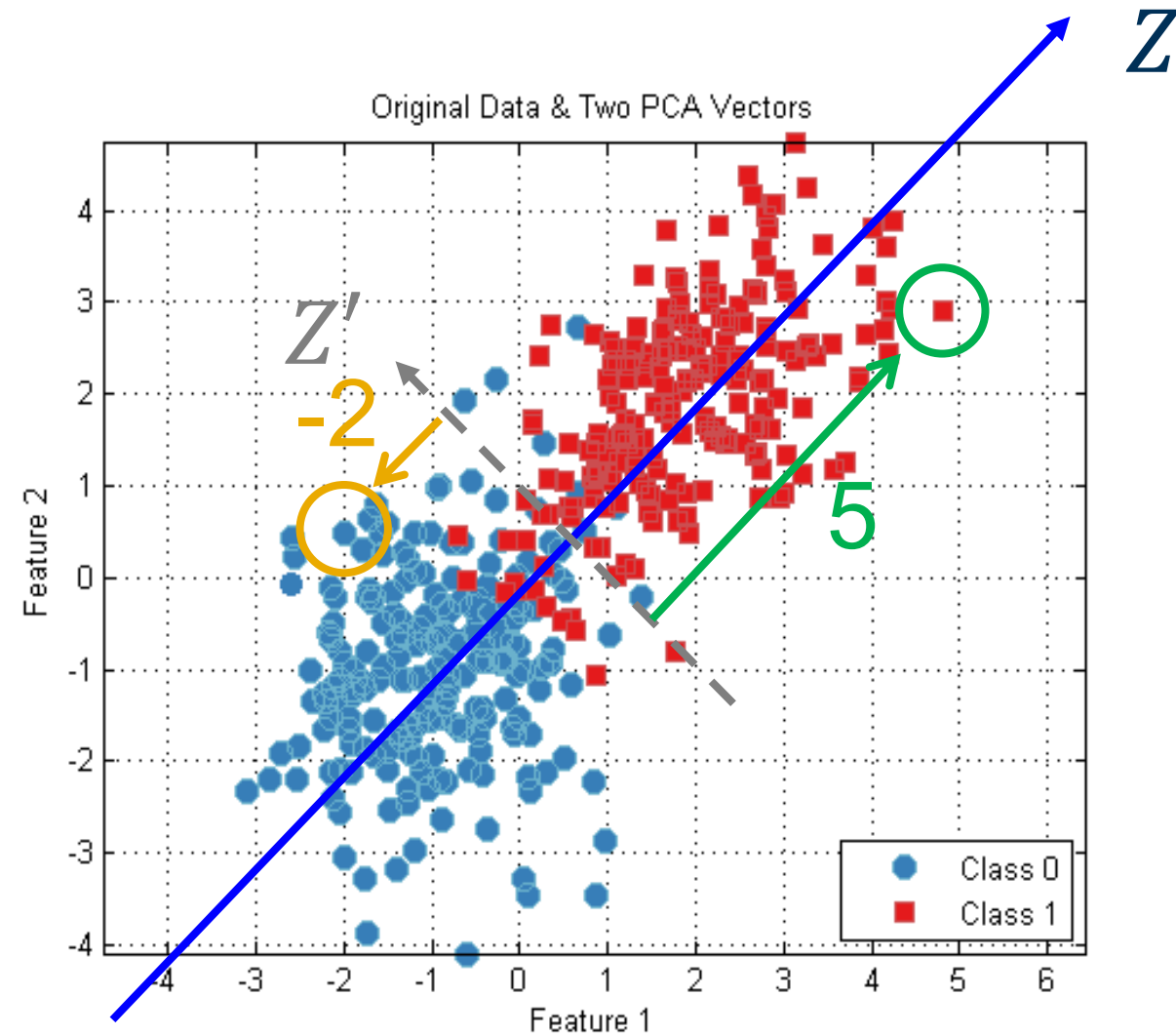
An Example



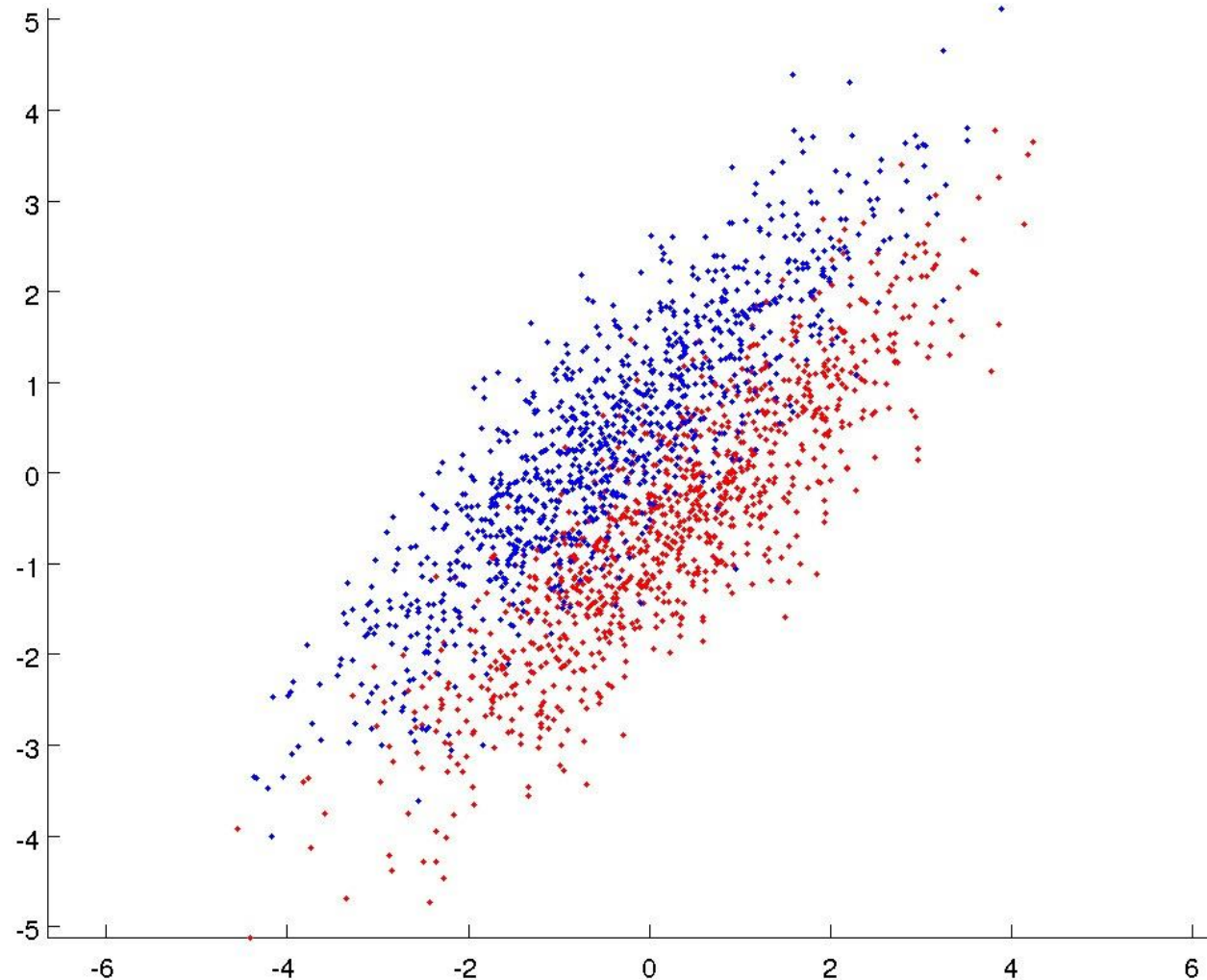
Principal Directions of the Data



Reduce to 1 Dimension



Are Principal Components Good for Classification?



When to Use PCA?

- **Visualization:** reduce dimension to 2 or 3 dimensions so that you can plot
- **Feature distribution:** analyze variance, mean, distribution, etc.
- **Feature engineering:** identify independent principal directions, reduce # of features (e.g., # features >> # data)
- **Data compression**
- **Drawbacks:**
 - Lose interpretability
 - Larger variance $\neq$ more information/predictability
 - Label agnostic, i.e., may not fit labels well

PCA on Leaves



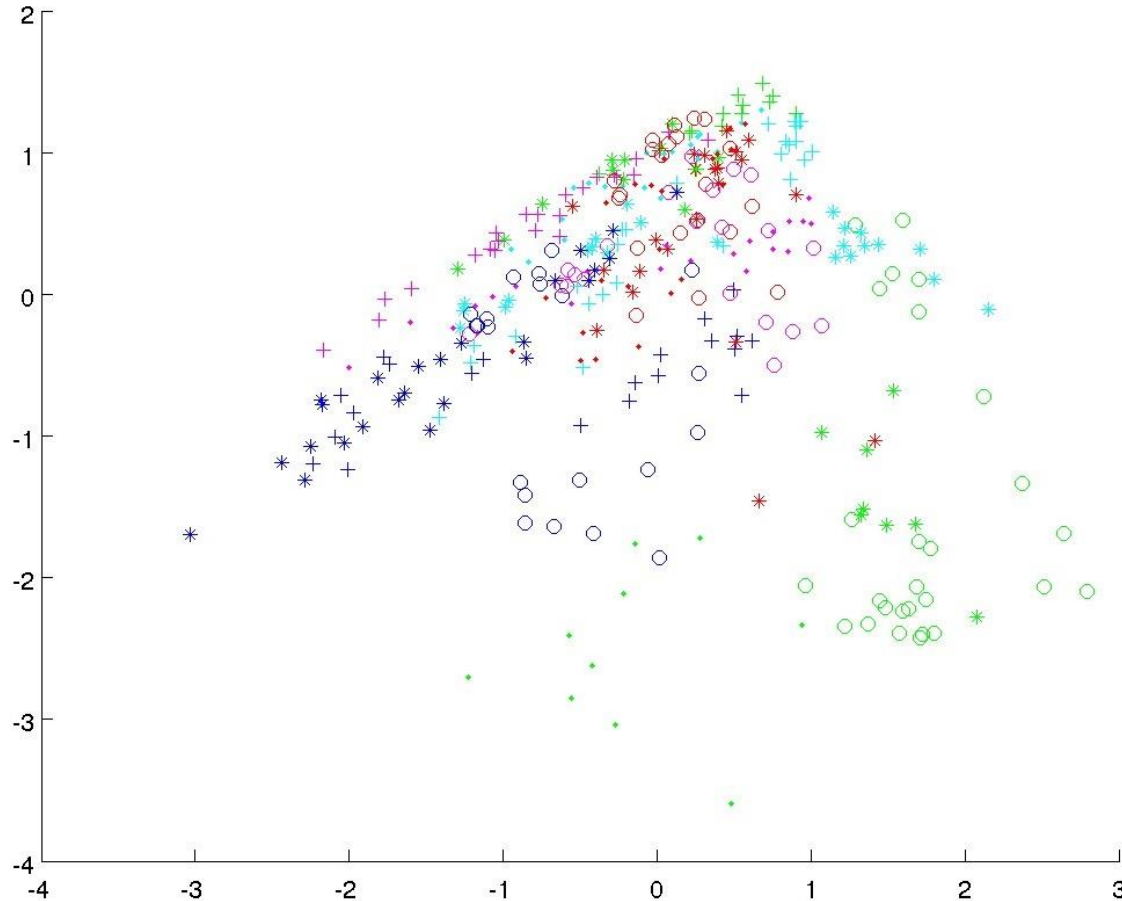
Input Features (representation)

Shape feature	Description
<i>Eccentricity</i>	Eccentricity of the ellipse with identical second moments to I . This value ranges from 0 to 1.
<i>Aspect Ratio</i>	Consider any $X, Y \in \partial I$. Choose X and Y such that $d(X, Y) = D(I)$. Find $Z, W \in \partial I$ maximizing $D^\perp = d(Z, W)$ on the set of all pairs of ∂I that define a segment orthogonal to $[XY]$. The aspect ratio is defined as the quotient $D(I)/D^\perp$. Values close to 0 indicate an elongated shape.
<i>Elongation</i>	Compute the maximum escape distance $d_{\max} = \max_{X \in I} d(X, \partial I)$. Elongation is obtained as $1 - 2d_{\max}/D(I)$ and ranges from 0 to 1. The minimum is achieved for a circular region. Note that the ratio $2d_{\max}/D(I)$ is the quotient between the diameter of the largest inscribed circle and the diameter of the smallest circumscribed circle.
<i>Solidity</i>	The ratio $A(I)/A(H(I))$ is computed, which can be understood as a certain measure of convexity. It measures how well I fits a convex shape.
<i>Stochastic Convexity</i>	This variable extends the usual notion of convexity in topological sense, using sampling to perform the calculation. The aim is to estimate the probability of a random segment $[XY]$, $X, Y \in I$, to be fully contained in I .
<i>Isoperimetric Factor</i>	The ratio $4\pi A(I)/L(\partial I)^2$ is calculated. The maximum value of 1 is reached for a circular region. Curvy intertwined contours yield low values.
<i>Maximal Indentation Depth</i>	Let $C_{H(I)}$ and $L(H(I))$ denote the centroid and arclength of $H(I)$. The distances $d(X, C_{H(I)})$ and $d(Y, C_{H(I)})$ are computed $\forall X \in H(I)$ and $\forall Y \in \partial I$. The indentation function can then be defined as $[d(X, C_{H(I)}) - d(Y, C_{H(I)})]/L(H(I))$, which is sampled at one degree intervals. The maximal indentation depth $\mathfrak{D}$ is the maximum of this function.
<i>Lobedness</i>	The Fourier Transform of the indentation function above is computed after mean removal. The resulting spectrum is normalized by the total energy. Calculate lobedness as $F \times \mathfrak{D}^2$, where F stands for the smallest frequency at which the cumulated energy exceeds 80%. This feature characterizes how lobed a leaf is.

Texture feature	Description
<i>Average Intensity</i>	Average intensity is defined as the mean of the intensity i
<i>Average Contrast</i>	Average contrast is the the standard deviation of the intensity, $\sigma = \sqrt{\mu_2(z)}$.
<i>Smoothness</i>	Smoothness is defined as $R = 1 - 1/(1 + \sigma^2)$ and measures the relative smoothness of the intensities in a given region. For a region of constant intensity, R takes the value 0 and R approaches 1 as regions exhibit larger disparities in intensity values. σ^2 is normalized by $(L - 1)^2$ to ensure that $R \in [0, 1]$.
<i>Third moment</i>	μ_3 is a measure of the intensity histogram's skewness. This is generally normalized by $(L - 1)^2$ like smoothness.
<i>Uniformity</i>	Defined as $U = \sum_{i=0}^{L-1} p^2(z_i)$, uniformity's maximum is reached when all intensity levels are equal.
<i>Entropy</i>	A measure of intensity randomness.

8 shape features
6 texture features

Reduce Representation



Principal directions:

w_1 w_2

0.0938 0.1924

0.1902 0.0253

0.2266 -0.1800

-0.1850 0.4084

-0.1600 0.3825

-0.2063 0.3488

0.1940 -0.4037

0.2150 -0.3566

Shape
features

Texture
features

-0.3723 -0.2001

-0.3657 -0.1974

-0.3602 -0.2037

-0.3175 -0.1886

-0.3056 -0.1243

-0.3482 -0.1829

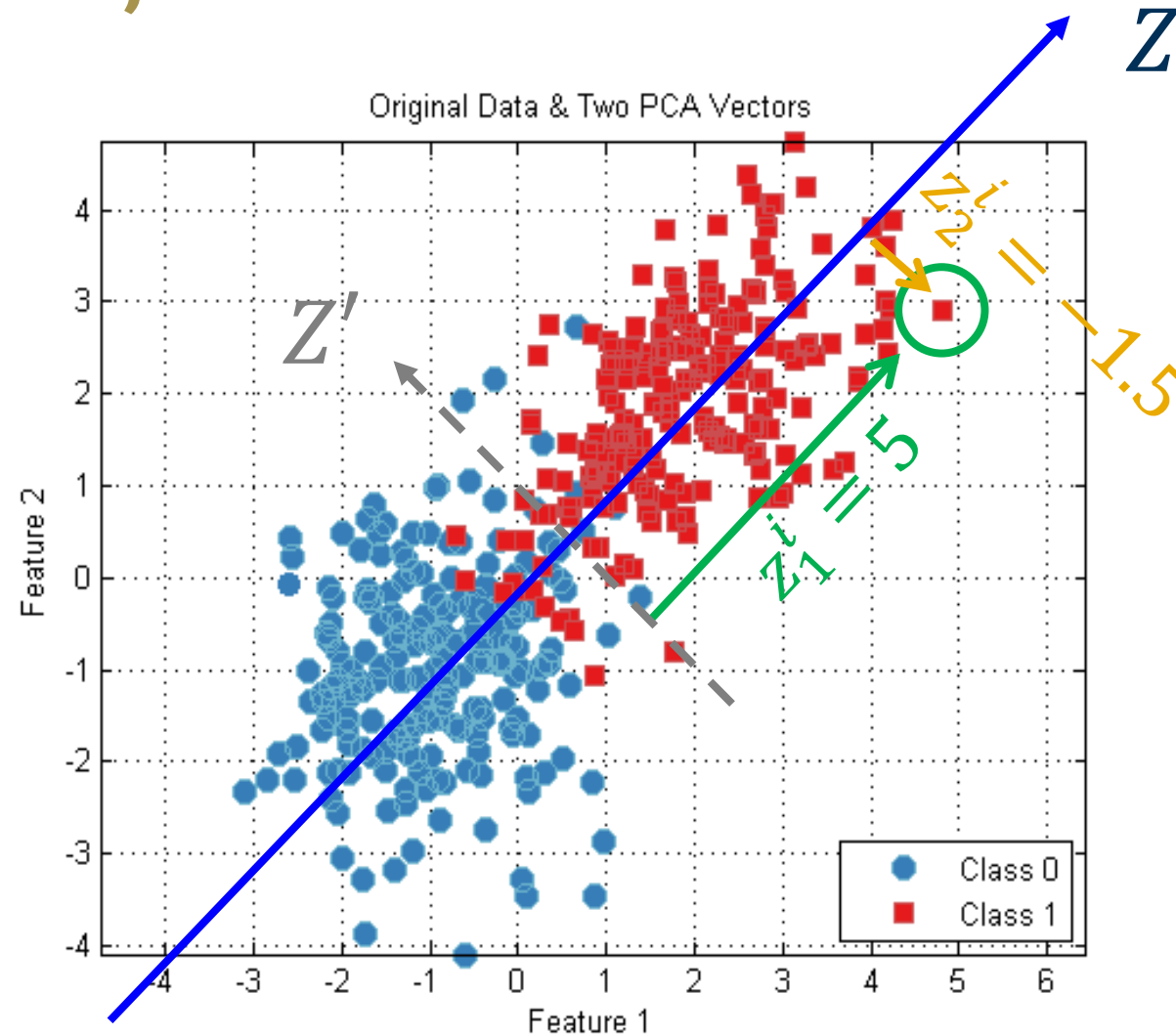
How to Recover the Original Data Point

- Given data mean μ , principal directions $w^1, w^2, \dots, w^d$, and the corresponding eigenvalues $\lambda_1, \lambda_2, \dots$
- Can we recover x^i from the reduced representation z^i **approximately**?

$$z^i = \begin{pmatrix} z_1^i \\ z_2^i \\ \dots \\ z_{d'}^i \end{pmatrix} = \begin{pmatrix} w^{1\top} (x^i - \mu) / \sqrt{\lambda_1} \\ w^{2\top} (x^i - \mu) / \sqrt{\lambda_2} \\ \dots \\ w^{d'\top} (x^i - \mu) / \sqrt{\lambda_{d'}} \end{pmatrix}$$

- $x^i \approx \hat{x}^i = \mu + z_1^i \cdot \sqrt{\lambda_1} \cdot w^1 + z_2^i \cdot \sqrt{\lambda_2} \cdot w^2 + \dots$

Reduce to 1-Dim, Reconstruct 2-Dim

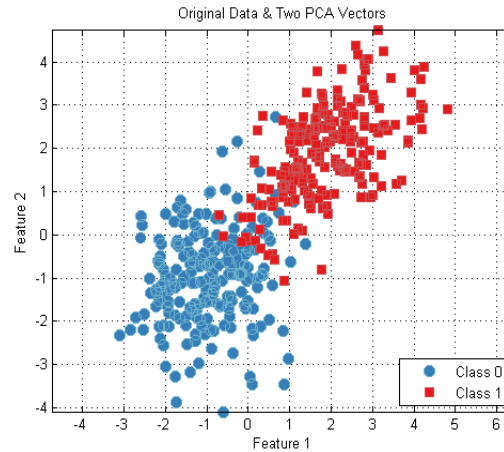


$$x^i \approx \hat{x}^i = \mu + \boxed{z_1^i \cdot \sqrt{\lambda_1} \cdot w^1} + \boxed{z_2^i \cdot \sqrt{\lambda_2} \cdot w^2} + \dots$$

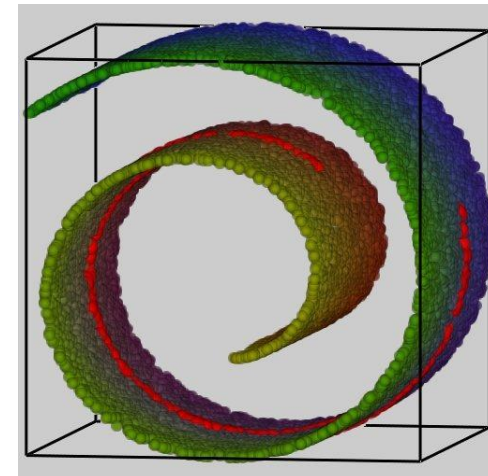
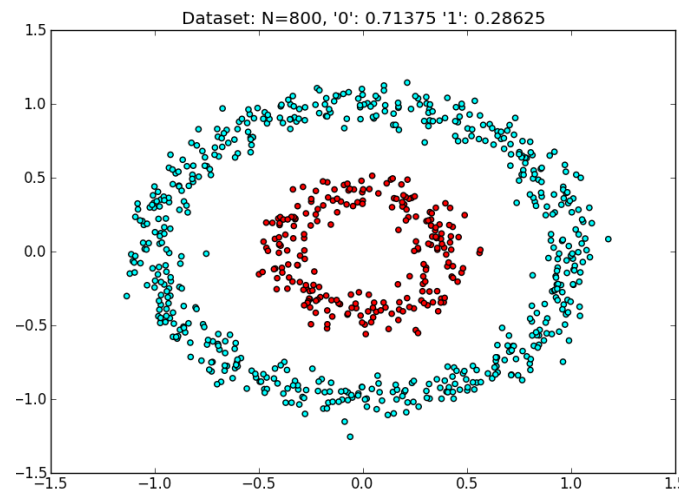
Non-linear Dimensionality Reduction

Limitation of PCA

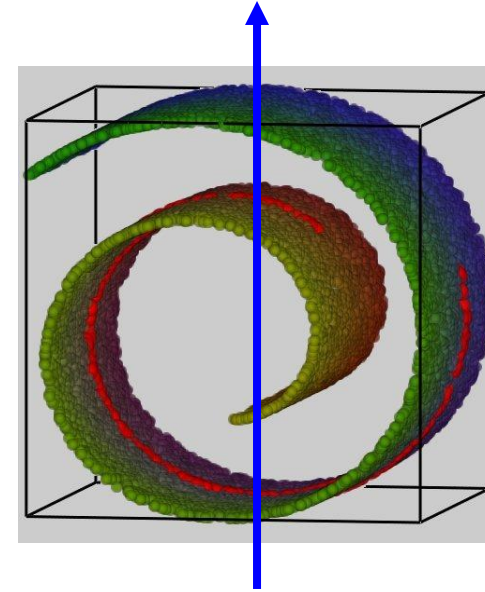
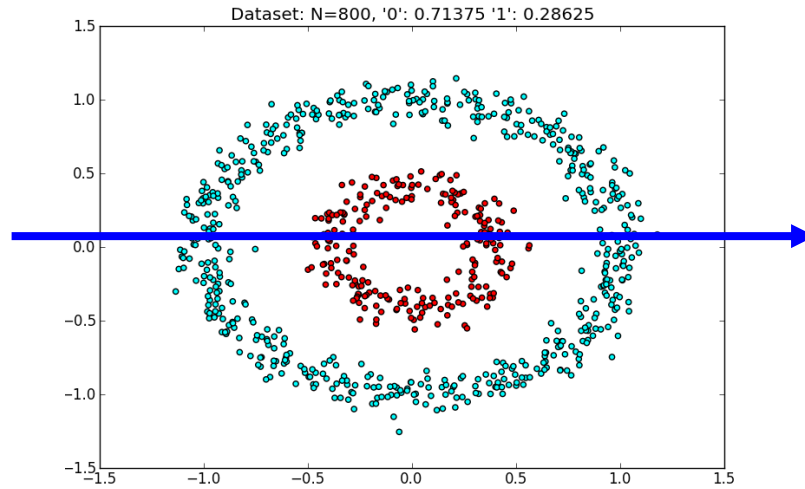
- Suitable when variables are linearly correlated



- Not suitable when nonlinear structures are present

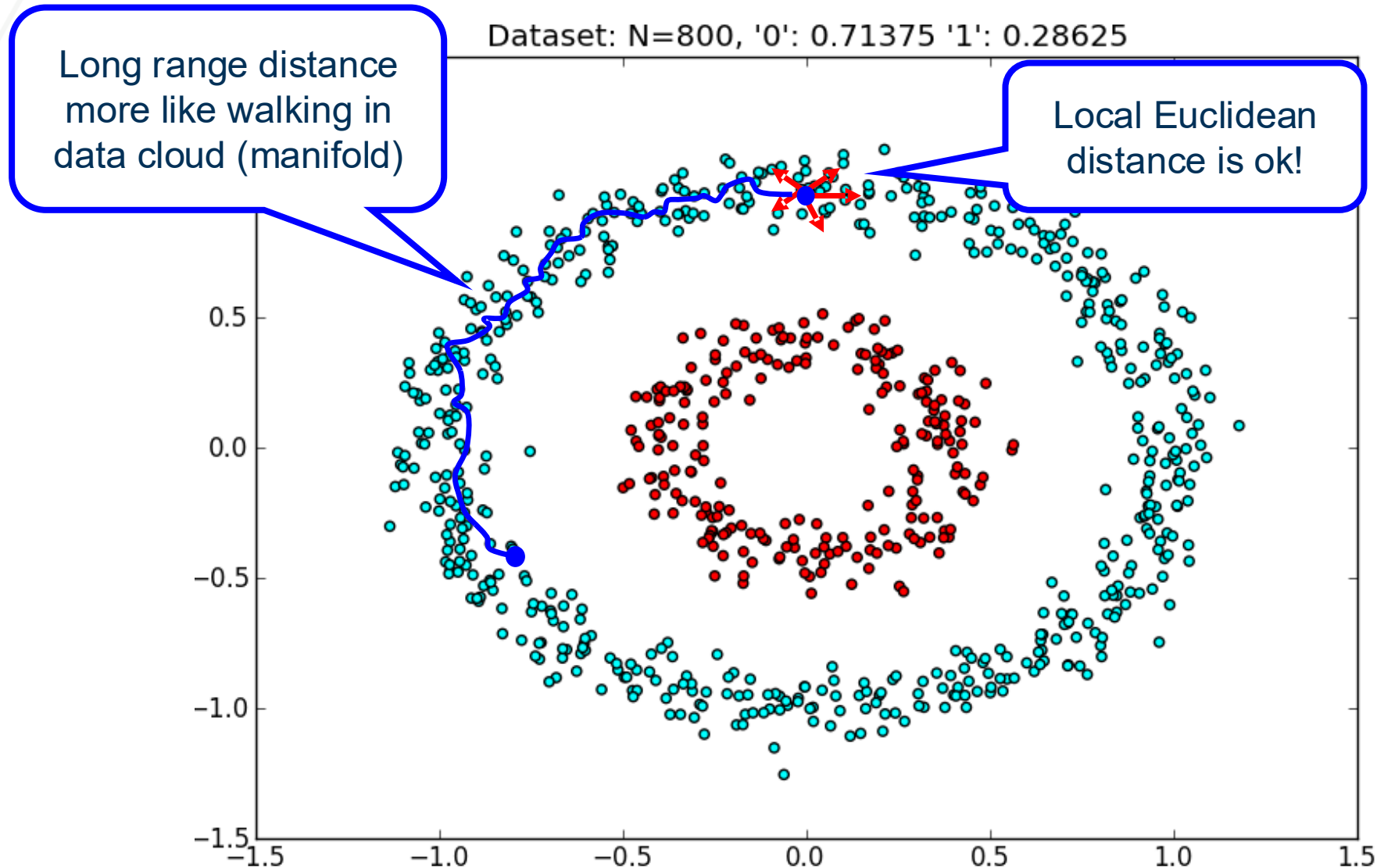


What's Wrong with PCA?



- PCA uses linear projection $w^T x$, implicitly assuming Euclidean distance is the dissimilarity (distance) measure
- When there are nonlinear structure, Euclidean distance is **not** the right distance measure **globally**.

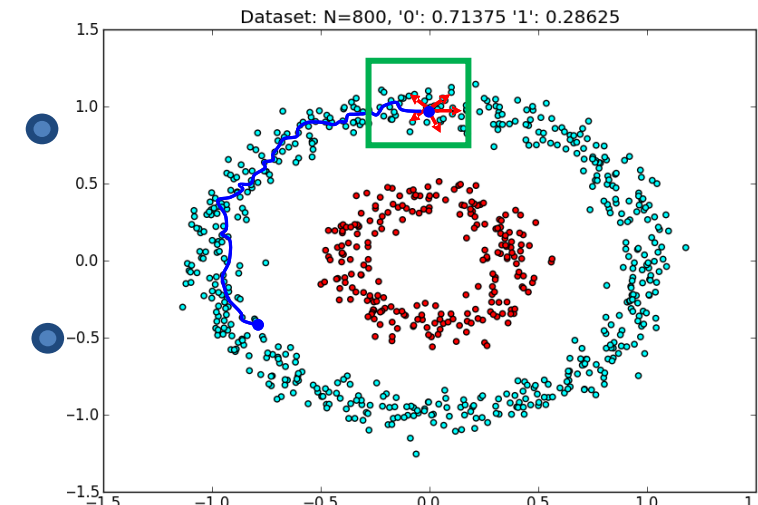
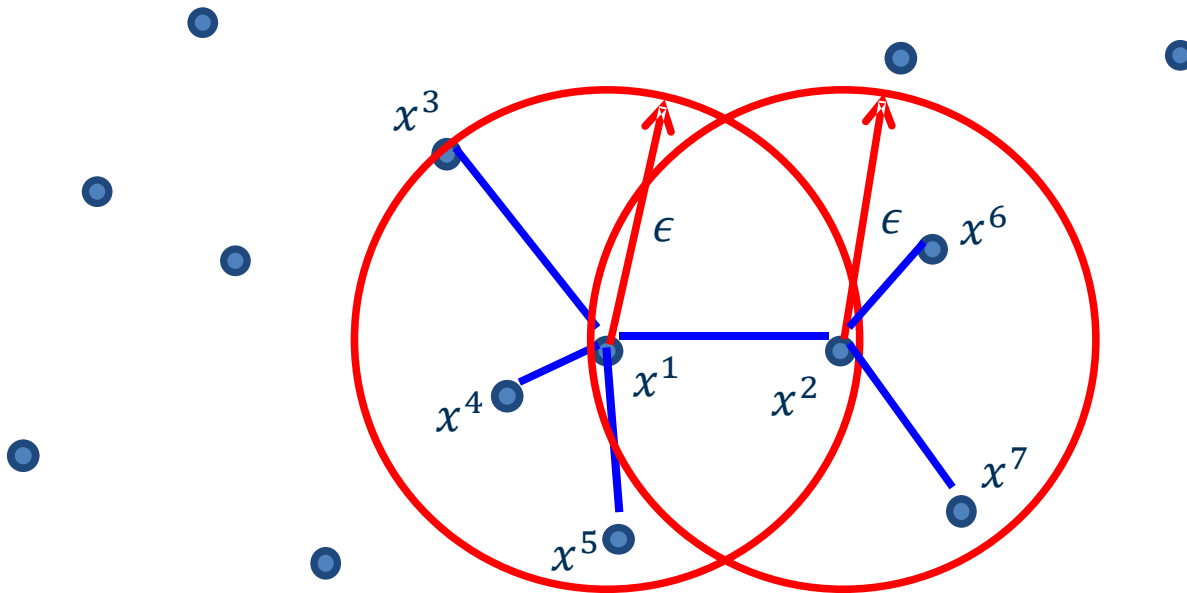
What's a Reasonable Distance Measure?



Using Neighbor Graph to Define Distance

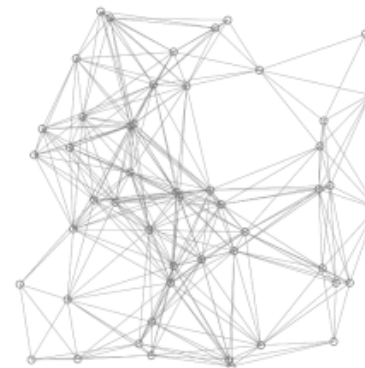
- Given n data points, threshold ϵ , construct adjacency matrix $A \in \mathbb{R}^{n \times n}$

$$A_{ij} = \begin{cases} 1, & \text{if } \|x^i - x^j\| \leq \epsilon \\ 0, & \text{otherwise} \end{cases}$$



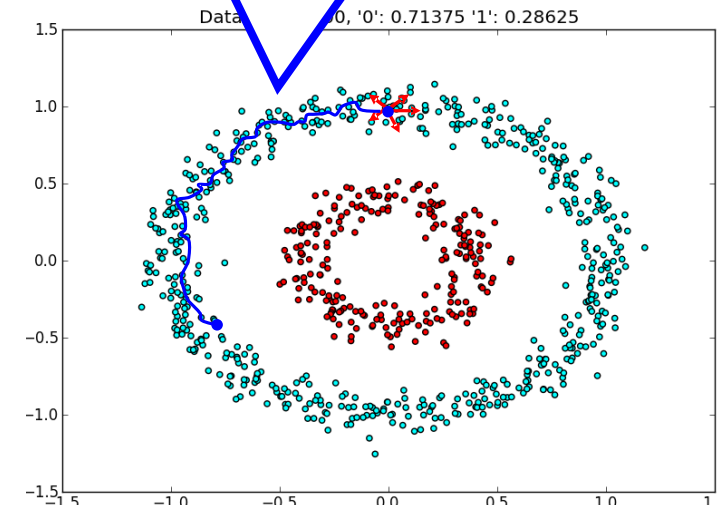
Shortest Path Distance

- With the graph defined by $A \in \mathbb{R}^{n \times n}$, find the shortest path distance matrix D between any pairs of points.
 - Aka. graph distance matrix
- The shortest path distance matrix D can be computed by:
 - Floyd-Warshall algorithm (all pair shortest path problem)
 - Cost: $O(|V|^3) = O(n^3)$
 - Dijkstra's algorithm * n
 - Cost: $O(n(|E| + |V| \log |V|)) = O(n|E| + n^2 \log n)$



Images from Wikipedia

How many steps to move from one point to another



From Distances to Reconstruct Representation

- **Goal:** Given the distance matrix D , find representation $z^i \in \mathbb{R}^{d'} \forall i$ such that

$$\begin{aligned} D_{ij}^2 &= \|z^i - z^j\|^2 \\ &= (z^i - z^j)^\top (z^i - z^j) \\ &= z^{i\top} z^i + z^{j\top} z^j - 2z^{i\top} z^j \end{aligned}$$

- In matrix format, let $Z = (z^1, z^2, \dots, z^n)^\top \in \mathbb{R}^{n \times d'}$

$$(D)^2 = a1^\top + 1a^\top - 2ZZ^\top \in \mathbb{R}^{n \times n}. \quad (\text{pairwise distance})$$

$$\text{where } a = (z^1^\top z^1, z^2^\top z^2, \dots, z^n^\top z^n)^\top$$

From Distances to Reconstruct Representation

- Construct a special centering matrix $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$
 - Verify
 - $\left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) \mathbf{1} \mathbf{a}^\top \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) = 0$
 - $\left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) \mathbf{a} \mathbf{1}^\top \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) = 0$
- Then apply H to both side of $(D)^2$
 - $C = -\frac{1}{2} H(D)^2 H = -\frac{1}{2} H(\mathbf{a} \mathbf{1}^\top + \mathbf{1} \mathbf{a}^\top - 2\mathbf{Z}\mathbf{Z}^\top) H = H\mathbf{Z}\mathbf{Z}^\top H$
 - $H\mathbf{Z} = \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\right) \mathbf{Z} = \mathbf{Z} - \mu \mathbf{1}^\top = \tilde{\mathbf{Z}}$
 - Ultimately we get $C = \tilde{\mathbf{Z}}\tilde{\mathbf{Z}}^\top$

Obtain Low-dimensional Representation

- Given $C = -\frac{1}{2}H(D)^2H = \tilde{Z}\tilde{Z}^\top$
- Perform eigenvalue decomposition on C
 - $Cw = \lambda w$
 - Take the eigenvectors $w^1, w^2, \dots$ of C corresponding to
 - The largest eigenvalue λ_1 , as the first coordinate
 - The second largest eigenvalue λ_2 , as the second coordinate...
- Reduced representation

$$\tilde{Z} = (w^1, w^2, \dots, w^{d'}) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \dots & \\ & & \lambda_{d'}^{1/2} \end{pmatrix}$$

Isomap

- Given n data points, $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- **Step 1:** build a weighted graph A using nearest neighbors, and compute pairwise shortest distance matrix D
- **Step 2:** use a centering matrix $H = I - \frac{1}{n} 11^\top$ to define covariance matrix

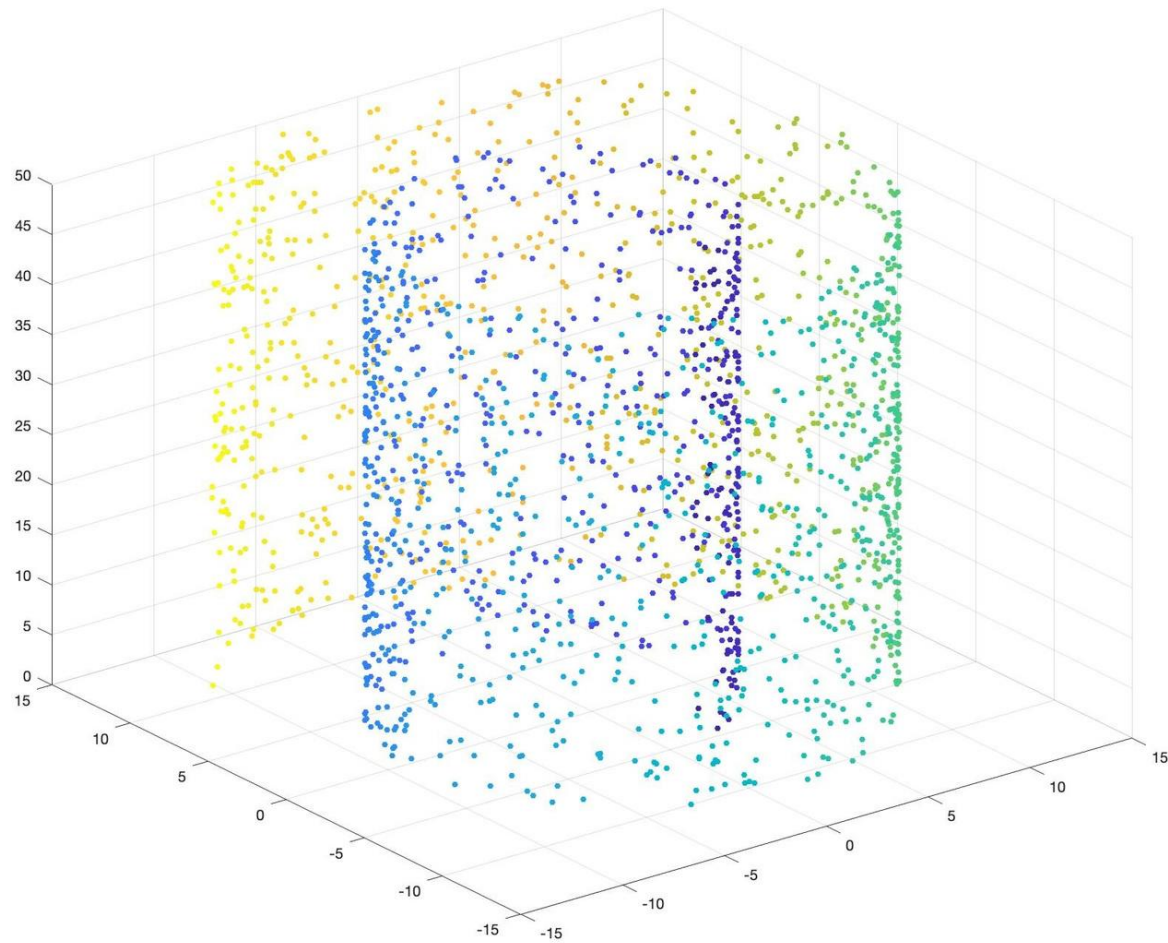
$$C = -\frac{1}{2} H(D)^2 H$$

Where $(D)^2 = (D_{ij}^2)_{i,j \in [1,2,\dots,n]}$

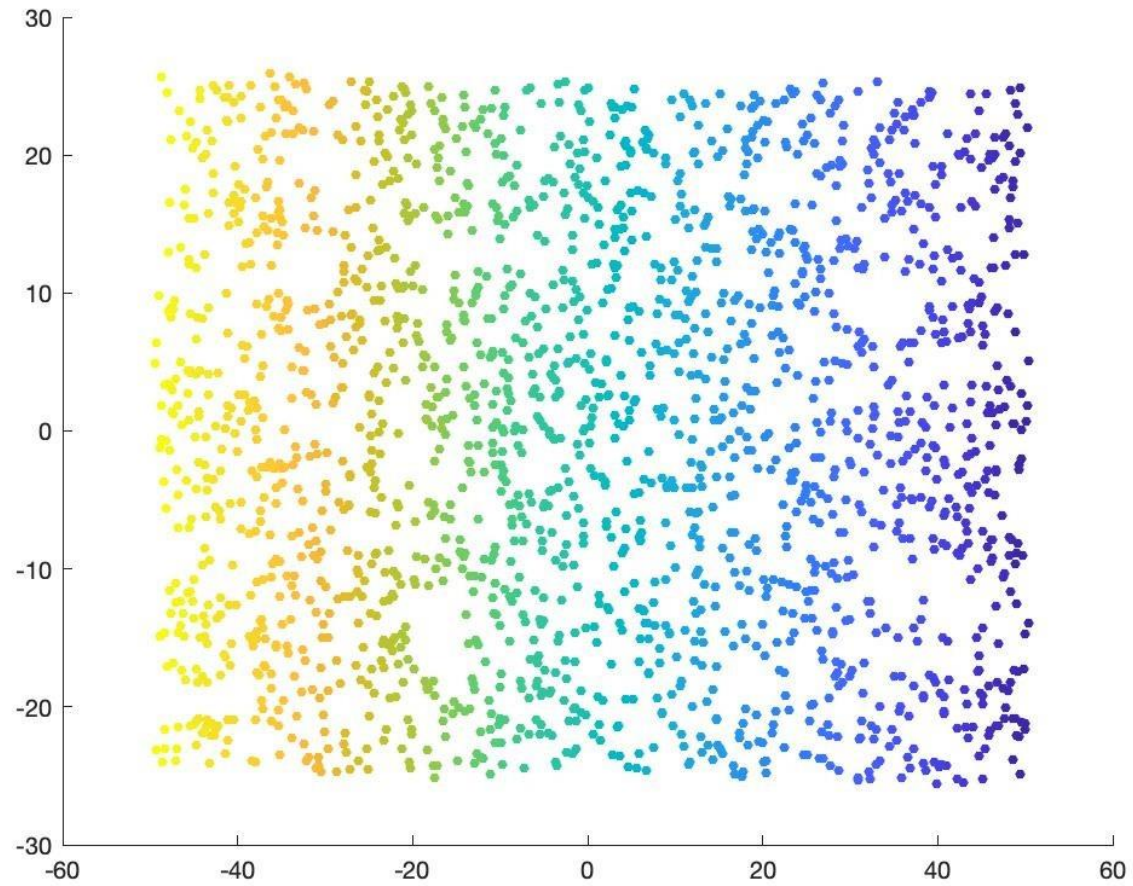
- **Step 3:** compute leading eigenvectors $w^1, w^2, \dots, w^{d'}$ and eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{d'}$ of C

$$\tilde{Z} = (w^1, w^2, \dots, w^{d'}) \begin{pmatrix} \lambda_1^{1/2} & & \\ & \dots & \\ & & \lambda_{d'}^{1/2} \end{pmatrix}$$

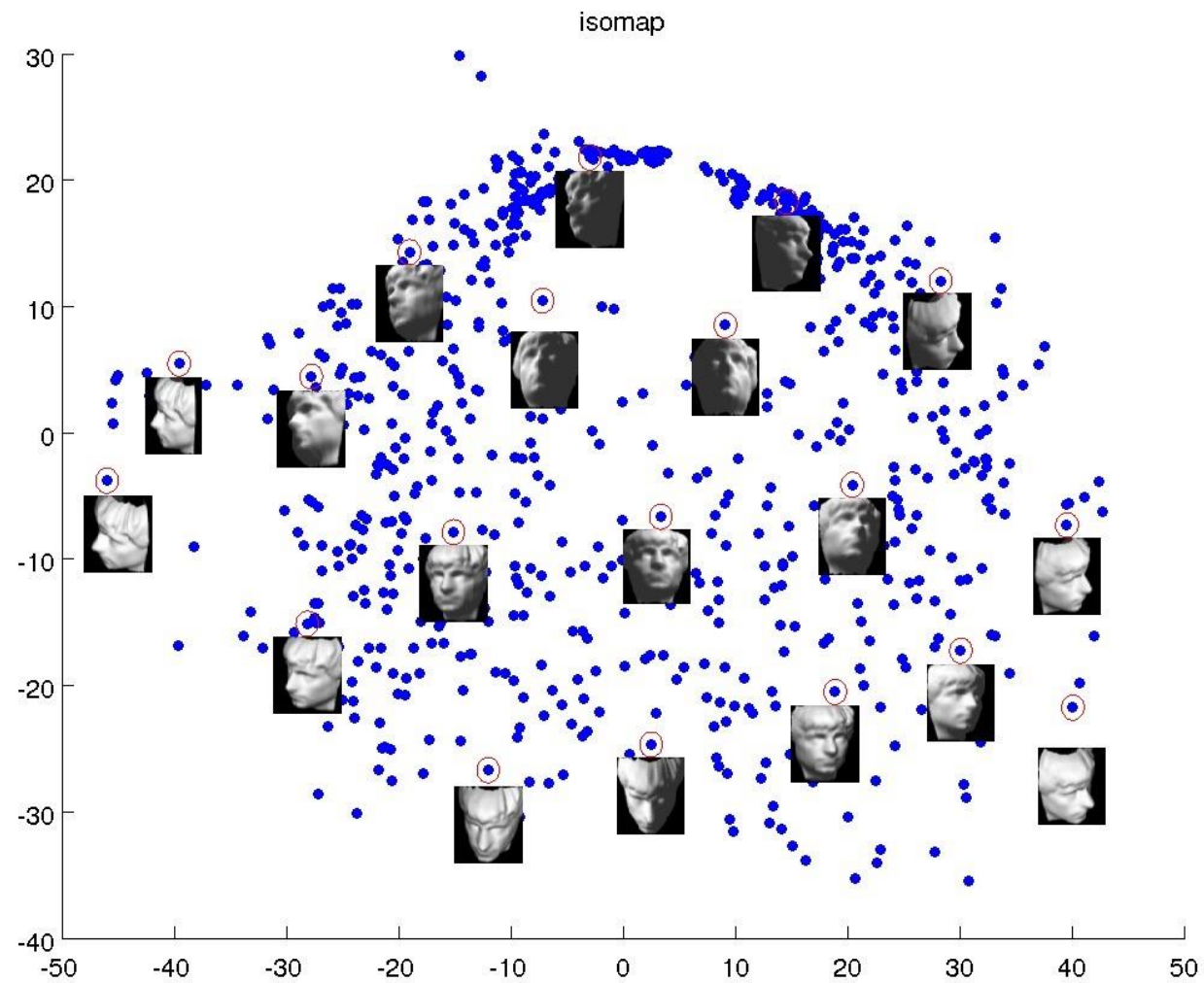
Swissroll



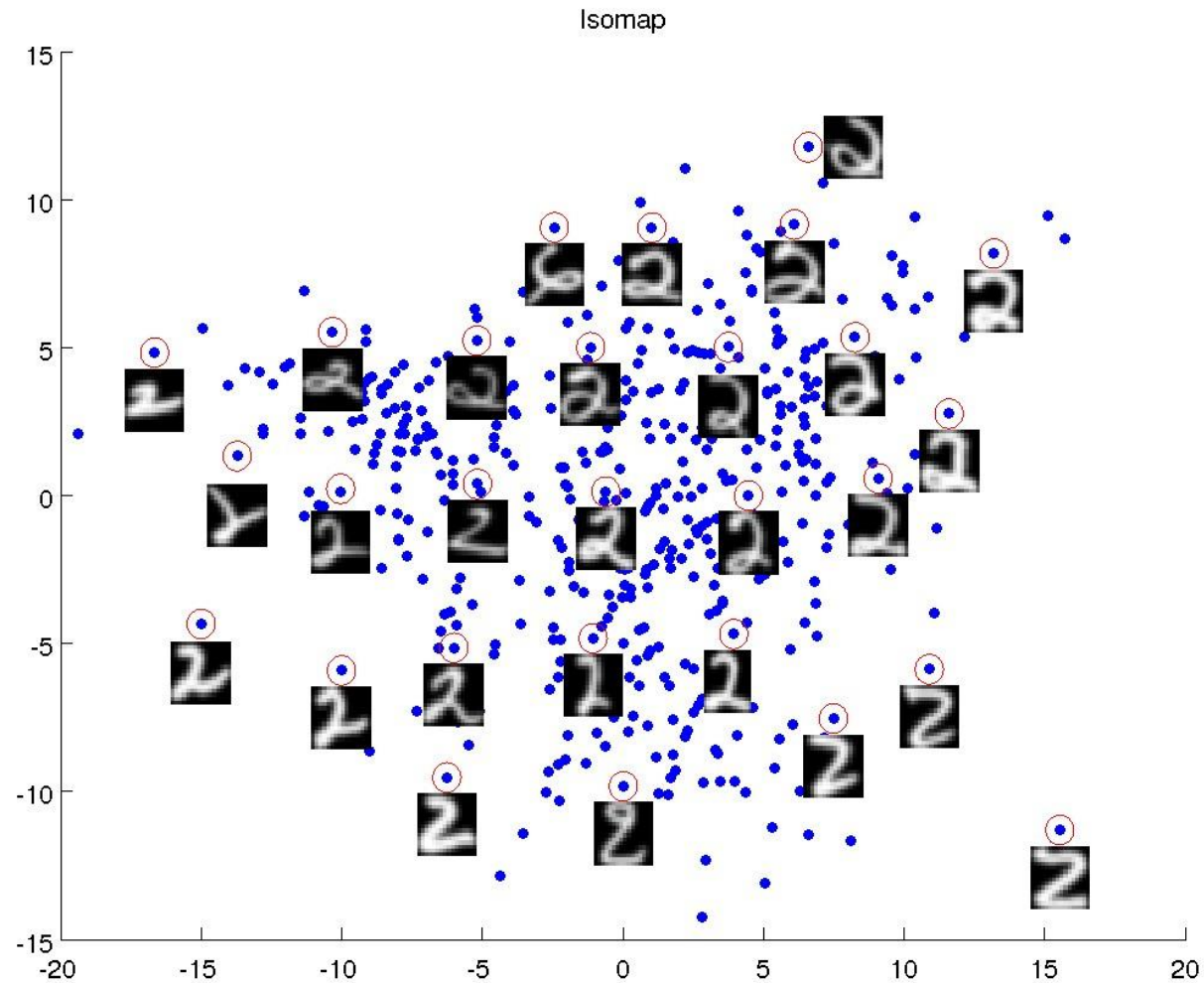
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Faces

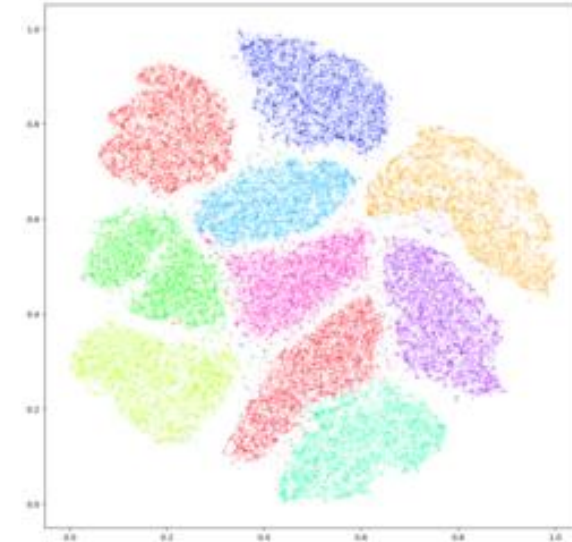


Handwritten Digits



Takeaway

- **PCA reduces dimensions by finding the top-k principal directions**
 - Principal directions are equivalent to the directions with largest eigenvalues of the covariance matrix
 - It is also equivalent to directions that maximize the variance
- **When to use dimensionality reduction?**
 - Feature distribution analysis, feature engineering
 - Visualization
 - Note that dimensionality reduction finds directions maximizing variance, but not the predictive power
- **When to use linear/non-linear dimensionality reduction?**
 - Linear/non-linear similarity/distance metric
 - Non-linear dimensionality reduction: [Isomap](#), SNE, [t-SNE](#), [kernel PCA](#)



T-SNE embeddings
of MNIST dataset